

Customer Shopping Behavior Analysis

End-to-End Data Analytics Portfolio Project

TOOLS & TECHNOLOGIES

 Python	 MySQL	 Power BI	 Pandas
 Jupyter Notebook	 NumPy	 Matplotlib	 SQL (DML/DQL)

1. Project Overview

This project is a complete end-to-end data analytics solution built to analyze customer shopping behavior using real-world transactional data. The workflow covers every stage of the analytics pipeline — from raw data ingestion and cleaning in Python, to structured SQL-based business querying in MySQL, through to an interactive executive dashboard in Power BI.

The primary objective is to uncover actionable insights into customer spending patterns, product category preferences, subscriber vs. non-subscriber behavior, and customer loyalty segmentation — ultimately providing data-driven recommendations that guide strategic business decisions.

2. Dataset Summary

Total Records	3,900 purchase transactions
Total Columns	18 feature columns
Missing Data	37 missing values in Review Rating column — imputed using category median
Data Source	Retail transactional dataset covering multiple product categories

Key Feature Groups

 Demographics	 Purchase Details	 Shopping Behavior
Age Gender Location Subscription Status	Item Purchased Category Purchase Amount Season · Size · Color	Discount Applied Previous Purchases Frequency of Purchases Review Rating · Shipping Type

 3. Exploratory Data Analysis (Python)

The data preparation and cleaning phase was conducted entirely in Python using pandas and NumPy. The following steps were applied systematically:

3.1 Data Loading & Initial Exploration

- **Data Loading** – Imported the full dataset using `pandas.read_csv()`, verified shape and column names.
- **Initial Exploration** – Used `df.info()` to inspect column dtypes and `df.describe()` to compute summary statistics for all numerical features.
- **Missing Data Handling** – Detected 37 null values in the Review Rating column. Imputed missing values using the median rating per product category to preserve distribution integrity.
- **Column Standardization** – Renamed all columns to snake case format for consistency across Python, SQL, and documentation.

3.2 Feature Engineering

- **Age_group** – Binned customer ages into meaningful segments (e.g., 18–25, 26–35, 36–50, 51+) using `pd.cut()` to enable demographic analysis.
- **Purchase_frequency_days** – Derived a numeric frequency column from the categorical purchase frequency data to support quantitative segmentation.
- **Data Consistency Check** – Investigated correlation between `discount_applied` and `promo_code_used`. Found them to be redundant; dropped `promo_code_used` to eliminate multicollinearity.

3.3 Database Integration

- **MySQL Connection** – Used SQLAlchemy to establish a database connection and load the cleaned pandas DataFrame directly into a MySQL database table.

- **Schema Design** – Defined appropriate column data types to ensure SQL query efficiency and data integrity.

4. Data Analysis using SQL

Structured business queries were executed in MySQL to answer ten key business questions. Each query was designed to translate directly into a business insight:

	Business Query	Insight Generated
▶	Revenue by Gender	Compared total revenue generated by male vs. female customers.
▶	High-Spending Discount Users	Identified customers who used discounts but still spent above the average purchase amount.
▶	Top 5 Products by Rating	Found products with the highest average review ratings.
▶	Shipping Type Comparison	Compared average purchase amounts between Standard and Express shipping.
▶	Subscribers vs. Non-Subscribers	Compared average spend and total revenue across subscription status.
▶	Discount-Dependent Products	Identified 5 products with the highest percentage of discounted purchases.
▶	Customer Segmentation	Classified customers into New, Returning, and Loyal segments based on purchase history.
▶	Top 3 Products per Category	Listed the most purchased products within each product category.
▶	Repeat Buyers & Subscriptions	Checked whether customers with >5 purchases are more likely to subscribe.
▶	Revenue by Age Group	Calculated total revenue contribution of each age group.

5. Dashboard in Power BI

An interactive executive dashboard was designed and built in Microsoft Power BI to visually present the insights derived from the SQL analysis. The dashboard is structured around the following key visualization panels:

- **Revenue Overview** – High-level KPI cards showing total revenue, average order value, and customer count with period-over-period comparison.
- **Customer Segmentation** – Donut and bar charts breaking down customers by New / Returning / Loyal segments and subscription status.
- **Demographic Analysis** – Gender-split revenue comparison and age group revenue contribution charts.
- **Product Performance** – Top-rated products, best-selling categories, and discount dependency analysis by product.
- **Shipping & Discounts** – Comparison of purchase behavior across shipping types and promotional usage patterns.
- **Seasonal Trends** – Revenue and purchase volume broken down by season to identify demand peaks.

The dashboard is designed with slicers for Season, Gender, Age Group, Subscription Status, and Category — enabling stakeholders to drill down and explore specific customer cohorts interactively.



6. Business Recommendations

Based on the findings from the SQL and Power BI analysis, the following strategic recommendations are proposed:

Boost Subscriptions

Data shows subscribers generate higher average revenue per transaction. Promote exclusive benefits — early access, free shipping, loyalty points — to convert non-subscribers and increase lifetime value.

Customer Loyalty Programs

Customers with 5+ purchases represent the highest-value segment. Introduce tiered rewards programs to move Returning customers into the Loyal segment and reduce churn.

Review Discount Policy

Several products show high discount dependency without proportional revenue uplift. Evaluate which discounts drive new customers vs. those that simply erode margin on existing buyers.

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns and storefront placements. Leverage review data as social proof in advertising.

Targeted Marketing by Demographics

Focus campaign spend on high-revenue age groups and Express-shipping users, who demonstrate higher willingness to spend. Personalize messaging by gender-based category preferences.

7. End-to-End Project Workflow



This document is part of a data analytics portfolio project.

Built with Python · MySQL · Power BI